Health Connect Rwanda Healthcare Analytics Data Quality Issues, Assumptions and Analytical Limitations

1. Patients Without Matching Appointment Records

Finding: The Patients dataset contains 20,000 records, while the Appointments dataset contains 5,000 records. A matching check identified 4,443 distinct patients with appointments and 15,557 patients without matching appointment records.

Potential impact: Appointment-based analysis may not represent all patients in the patient database.

Limitation: The reason these patients lack matching appointments is unknown. They may be registered patients without appointments in the available data, or their appointment records may be missing.

Action: Treat patients without matching appointments as having no linked appointment in the available dataset. Do not assume they never received healthcare services.

2. Unmatched Appointment IDs Across Clinical and Operational Datasets

Finding: Some AppointmentID values in Laboratory Tests, Prescriptions, Admissions, Referrals, Patient Feedback, and Billing do not exist in the Appointments dataset.

Dataset	Total records
Laboratory Tests	5,000
Prescriptions	5,000
Admissions	5,000
Referrals	4,709
Patient Feedback	21,064
Billing	60,000

Potential impact: Records with unmatched appointment IDs cannot reliably be linked to appointment details, potentially affecting analysis by facility, department, patient, or appointment date.

Possible explanation: The pattern suggests that some appointment records referenced by other datasets may be missing from the available Appointments dataset. However, the cause has not been confirmed.

Action: Quantify unmatched records in each dataset and investigate them against the source data. Do not automatically delete them. Where appropriate, conduct analysis using matched records and disclose the resulting limitation.

3. Billing and Revenue Summary Appointment Count Discrepancy

Finding: The Billing dataset contains 60,000 records and 60,000 distinct AppointmentID values. Revenue Summary reports an appointment total of 60,809.

Difference: Revenue Summary reports 809 more appointments than the number of distinct appointment IDs in Billing.

Potential impact: Using the figures interchangeably may lead to inaccurate reporting of appointment volume or revenue-related metrics.

Limitation: The reason for the discrepancy is not established. The datasets may use different reporting definitions, data coverage, or counting methods.

Action: Validate how Appointment Total is calculated in Revenue Summary and confirm that both datasets cover the same reporting period and use comparable definitions before reconciling the figures.

4. Appointment Count Difference Between Appointments and Billing

Finding: The Appointments dataset contains 5,000 records, while Billing contains 60,000 distinct appointment IDs.

Potential impact: This substantial difference suggests that the available Appointments dataset may not contain all appointment records referenced in Billing.

Limitation: The cause is not confirmed. The difference should not automatically be interpreted as proof that appointment records were never entered.

Action: Investigate the unmatched IDs and establish whether the Appointments dataset is incomplete, has different data coverage, or uses a different identifier structure.

5. Financial Analysis Limitations

Finding: Billing contains `Amount` and `Status` fields, with statuses of Pending, Paid, and Partially Paid. Insurance Claims contains BillingID and Claim Amount, while Payments contains BillingID and `Amount Paid`.

Reported totals:

- Total billed amount: 9,309,401,961

- Total insurance claim amount: 4,881,863,294

- Total amount paid: 8,389,454,556

Potential impact: Financial analyses may be misleading if billing, claims, and payments are combined without validating their relationships and business rules.

Limitation: The totals alone do not establish how much remains outstanding, whether claims have been approved, or whether payments and claims overlap.

Action: Validate BillingID relationships across the three datasets. Confirm the meaning of claim amounts, payment statuses, and partial payments before calculating outstanding balances or comparing total billed, claimed, and paid amounts.

6. Patient Feedback Analysis Limitation

Finding: Patient Feedback contains 21,064 records, and some of its appointment IDs do not match the Appointments dataset.

Potential impact: Feedback records with unmatched appointment IDs cannot reliably be associated with appointment details, limiting satisfaction analysis by facility, department, or date.

Action: Quantify matched and unmatched feedback records. Analyze overall ratings and comments where appropriate, but qualify analyses that require appointment-level information.

7. General Validation Principles

Before finalizing the analysis:

1. Verify primary keys and foreign key relationships across datasets.
2. Quantify unmatched records rather than describing them only as “some.”
3. Distinguish missing relationships from missing data and avoid assuming the cause without evidence.
4. Validate totals and metric definitions before reconciling datasets.
5. Document exclusions, assumptions, and limitations for every analysis affected by data quality issues.
6. Retain valid records with unresolved relationships where possible, and avoid silently deleting data.

Overall Conclusion

The main data quality concern is the presence of appointment IDs in several datasets that cannot be matched to the available Appointments dataset. This may limit integrated clinical, operational, and financial analysis. A separate discrepancy exists between the 60,000 distinct appointment IDs in Billing and the 60,809 appointments reported in Revenue Summary.

These issues do not prevent all analysis. They require validation, careful selection of records, and transparent reporting of limitations before drawing conclusions.